

Nguyen Thanh Minh Duy

AI Engineer / ML Engineer

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Professional Summary

AI Engineer / ML Engineer with 4 years of experience across production software engineering, applied AI systems, and deep learning research. Former Software Engineer at Zalo with experience shipping production features and optimizing performance. Recent Master's graduate with Q1 publication in deep learning for intrusion detection, plus hands-on AI engineering projects in LLM applications, RAG, multimodal retrieval, FAISS, FastAPI, Docker, and PyTorch.

Education

Soonchunhyang University (SCH)

M.S. in Engineering (Fully Funded); **GPA: 4.32/4.5**

South Korea

Mar 2024 – Feb 2026

- **Focus:** Deep Learning, Few-Shot Learning, Open-Set Recognition, GNNs, Transformer-based anomaly detection.

HCM VNU – University of Science

B.Sc. in Software Engineering; **GPA: 8.66/10**

Ho Chi Minh City, Vietnam

Aug 2018 – Feb 2022

Technical Skills

- **AI Engineering:** LLM Applications, RAG, Agentic Workflows, Structured Outputs, Prompt Engineering, FAISS
- **ML / Deep Learning:** PyTorch, PyTorch Geometric, Transformers, GNNs, Few-Shot Learning, Open-Set Recognition, Anomaly Detection, CLIP/SigLIP
- **Backend / Deployment:** FastAPI, Docker, REST APIs, Pydantic, Git, API Integration
- **Frontend / Product:** Next.js, React.js, Electron.js, TypeScript
- **Languages:** Python, TypeScript/JavaScript

AI Engineering Projects

Bé Đậu – AI Career SaaS | AI + Full-Stack Engineer

2025 – Present

- Built an AI career platform for the Vietnamese market, including ATS-style CV analysis, job-fit evaluation, CV tailoring, and real-time voice mock interview practice.
- Designed a FastAPI backend for LLM-powered workflows using structured outputs and validation logic to make AI responses easier to parse, control, and integrate into user-facing dashboards.
- Implemented a multi-provider LLM fallback system using local/self-hosted inference with llama.cpp + Qwen, Gemini, and Groq, reducing dependency on paid APIs and improving resilience when providers are unavailable.
- Developed prompt workflows for HR, technical, and managerial interview personas, and built the Next.js product experience for CV analysis, AI feedback, interview simulation, and job-preparation workflows.

Interactive Video Retrieval System | AI + Fullstack Developer, Team of 4

Aug 2025 – Oct 2025

- Built a multimodal video retrieval system for competition-scale search across 200GB of video data, 800+ videos, and 1,431,212 extracted keyframes.
- Implemented CLIP/SigLIP-based text-to-keyframe retrieval with FAISS vector search, achieving sub-second retrieval latency for competition use cases.
- Contributed to AI core development through OCR-based data enrichment experiments, preprocessing workflows with ffmpeg and TransNet, and large-scale keyframe indexing.
- Built the React-based search interface for query input, result visualization, rapid keyframe inspection, and was selected as a finalist in the HCMC AI Competition.

AI Support Chatbot for Small Academic Center | Freelance AI Engineer

2025

- Built a FastAPI-based AI chatbot integrated with Facebook and Instagram webhooks to automate responses to course, pricing, and enrollment inquiries.
- Implemented a RAG pipeline using web crawling, OpenAI embeddings, ChromaDB vector search, and parent-child chunking to retrieve relevant academic center information.
- Designed session-aware conversation handling with prompt constraints and a validator LLM layer to preserve context, enforce approved responses, and reduce hallucinated answers.

Research Experience

Soonchunhyang University – IoT Network Lab

Graduate Research Assistant

Asan, South Korea

Feb 2024 – Feb 2026

- Researched deep learning approaches for in-vehicle CAN bus intrusion detection, focusing on data-scarce, imbalanced, few-shot, and open-set attack scenarios.
- First paper: developed a few-shot intrusion detection approach using metric learning, episodic training, and prototype-based classification to improve generalization to test-time attack classes.
- Achieved state-of-the-art multi-class classification results on the public CHD + ROAD CAN intrusion datasets while using less training data than conventional supervised approaches.
- Second paper: proposed an open-set CAN intrusion detection framework for both multi-class known attack classification and unseen attack detection, addressing a less-explored setting in in-vehicle network security.
- Designed a GNN-based model using node reconstruction loss, edge reconstruction loss, and classification/objective loss to capture structural anomaly signals from CAN message relationships.
- Achieved state-of-the-art known-attack classification performance while reaching 95% detection rate on unseen attack classes.
- Built GPU-based PyTorch / PyTorch Geometric training, validation, and evaluation pipelines for repeated experiments, class-wise metric analysis, ablation studies, and model comparison.
- Produced weekly research reports with visualizations, failure-case analysis, and experiment summaries; contributed to a Q1 Knowledge-Based Systems publication and a second paper currently under review.

Software Engineering Experience

Zalo – VNG Corporation

Software Engineer, Zalo PC

Ho Chi Minh City, Vietnam

May 2022 – Mar 2024

- Shipped 15+ production features for Zalo PC across messaging, contact, and desktop user experience surfaces.
- Optimized client-side message search using batching and caching, reducing search latency by 30% while preserving UI responsiveness across varied local database sizes.
- Led frontend implementation for a full architectural revamp of the user Contact Card, improving maintainability in a complex React/Electron legacy codebase.
- Maintained and extended large-scale TypeScript/JavaScript codebases used in production desktop software.
- Collaborated closely with Product, Design, Backend, and QA teams to translate ambiguous requirements into stable production releases.
- Debugged UI, state-management, and performance issues across legacy and newly developed Zalo PC features.
- Promoted from Fresher to Software Engineer due to consistent delivery quality, ownership, and production-readiness.

Publications

- **Nguyen T. M. Duy**, et al. “Multi-Class Intrusion Detection System for In-Vehicle Networks Using Few-Shot Learning and Convolutional Anomaly Transformer.” **Knowledge-Based Systems**, Q1 Journal, Nov 2025.
- “Open-Set Recognition with Multi-Objective Graph Neural Network in CAN.” Under review, 2025.

Certifications

IBM – Generative AI Engineering with LLMs Specialization

Mar 2026